

The Many Ends of Programming

Ray Myers
All Hands AI

This is not an AI talk



This is not a
programming talk



This is a talk about
empathy

This is a talk about
listening

A day in the life of
an AI skeptic



The Future of U.S. AI Leadership with CEO of Anthropic Dario Amodei



Share

Anthropic's CEO on the Future of AI

Watch on  YouTube



“

In 12 months we may be in a
world where AI is writing
essentially all the code

”



Ray Myers • You
Tech Lead | Mender | Untangler
3d • Edited • 🔄



A friendly challenge to [Anthropic](#), who's CEO predicted yesterday in a Council on Foreign Relations interview that it would be 12 months until the end of programming by humans.

If you intend those words to be taken seriously, I'll oblige. Mind you I use Claude and I'm a fan of their work, but...

Kill one (1) mainframe and report back with your updated timeline. Explain how you achieved it successfully without service disruption. Or have your AI explain if you like.

If I understand correctly that's orders of magnitude easier than what was claimed, and is a challenge relevant to companies that make up 50% of US GDP.

If that sounds too hard, here's a baby step:

Rewrite the TeX typesetting command in idiomatic Rust. It should be a drop-in replacement that produces the same output. In my understanding the current codebase is under 40K lines, has changed little since 1982, and is very well documented.

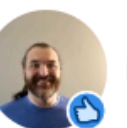
Again, follow up explaining the process and amount of human intervention required.

Should be easy, right?

Tim O'Farrell and 161 others

71 comments • 12 reposts

Reactions





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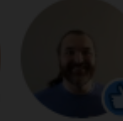
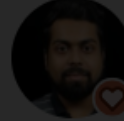
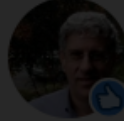
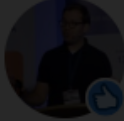
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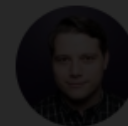
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Challenge 1

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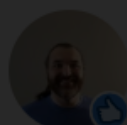
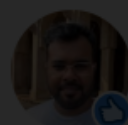
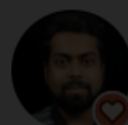
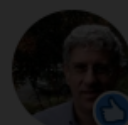
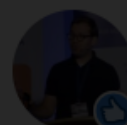
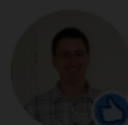
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“LLMs break old code”

- me, a lot

“Luddite”

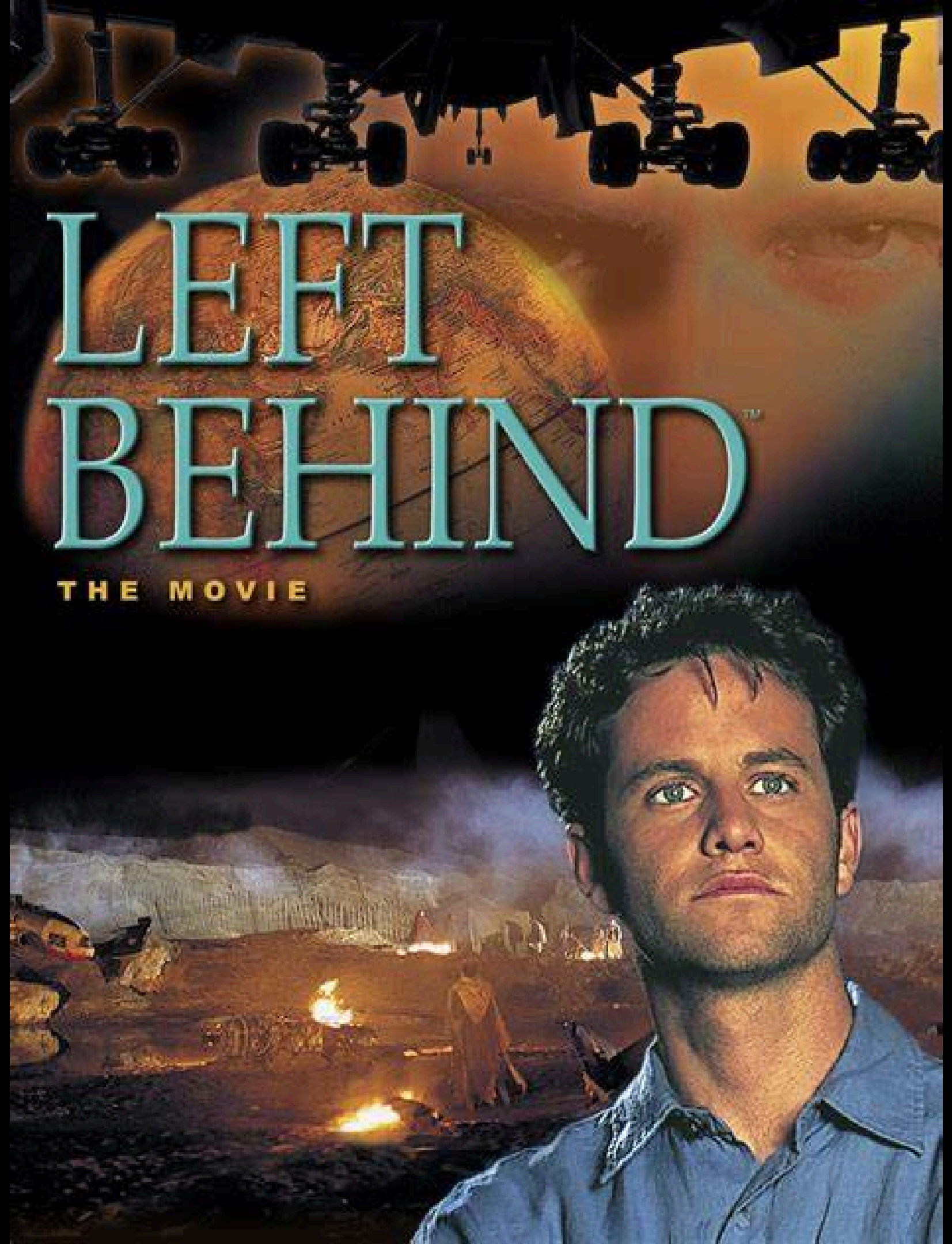
“You’ve got your head in the sand”

“Missing the big picture”

- True believers

“You are so completely missing the
big picture that it physically hurts
to read this post”

“Left behind”



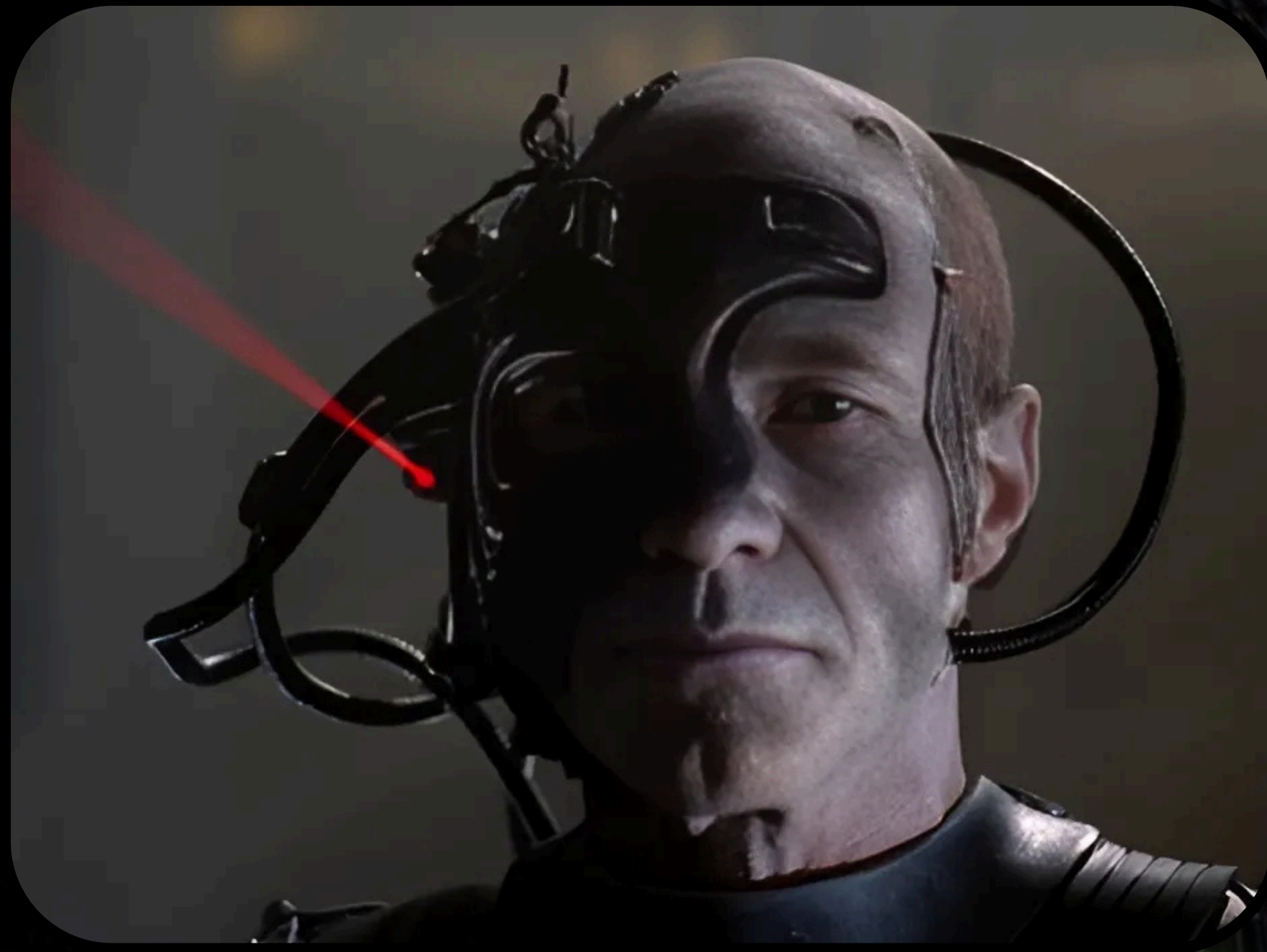
LEFT BEHIND™

THE MOVIE

“Resistance is futile”



Resistance is futile



Resistance is futile

Breath, and listen

What are we even
talking about?

What is the future of
software?

Six Scenarios

1. Extreme Completion
2. Dev-o-calyipse
3. Abstraction Leap
4. Uncharted Waters
5. Review Economy
6. Infinite Pile of Garbage

Extreme Completion

Extreme Completion

AI does the typing

Our job doesn't fundamentally change

Example: Code completion in Cursor

```
-- remove trailing whitespace  
clean' = map (reverse . dropWhile isSpace . reverse)
```

create clean" to remove leading and trailing spaces

 **Generate**

 quick question

 claude-3.7-sonnet Ctrl+Shift+K to toggle

Example: Code completion in Cursor

```
-- remove trailing whitespace
clean' = map (reverse . dropWhile isSpace . reverse)

-- remove both leading and trailing whitespace from each line
clean'' = map (reverse . dropWhile isSpace . reverse . dropWhile isSpace)
```

Example: OpenHands agent PR from Slack



Ray Myers Today at 4:13 PM

[@OpenHandsStaging](#) OpenHands PR [#8457](#) added exc_info=True to a log statement in an except block. There are probably others that should have something like that but don't. Could you prepare a PR that adds it to the appropriate places? You can ignore evaluation, just the main [openhands](#) folder

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! replies

OpenHandsStaging APP Today at 4:14 PM
I'm on it!

OpenHandsStaging APP Today at 4:26 PM

- Here's a summary of what was accomplished:
- Created and executed two utility scripts:
 - `find_log_in_except.py`: Identifies logger statements in except blocks missing `exc_info=True`
 - `add_exc_info.py`: Automatically adds `exc_info=True` to appropriate logger statements
- Modified 48 Python files in the `openhands` directory (excluding evaluation):
 - Added `exc_info=True` to `logger.error`, `logger.warning`, and `logger.critical` calls in except blocks
 - Fixed syntax errors in several files where the parameter was incorrectly added
- Created a new branch `add-exc-info-to-log-statements` and committed all changes:
 - The commit includes all modified files and the utility scripts
 - The branch has been pushed to the remote repository
- The PR is ready to be created manually through the GitHub web interface:
 - URL: <https://github.com/All-Hands-AI/OpenHands/pull/new/add-exc-info-to-log-statements>
 - Suggested title: "Add exc_info=True to logger statements in except blocks"
 - Suggested description: "This PR adds `exc_info=True` to logger statements in except blocks, similar to PR #8457. This ensures that exception tracebacks are properly logged when errors occur, making debugging easier."

The changes ensure that when exceptions occur, the full traceback information is logged, making debugging easier. This is a maintenance improvement that follows the pattern established in PR #8457.

Conversation 0 Commits 2 Checks 9 Files changed 48

Changes from all commits File filter Conversations

Filter changed files

openhands

- agenthub
 - browsing_agent
 - browsing_agent.py
 - response_parser.py
 - visualbrowsing_agent
 - visualbrowsing_agent.py
- cli
 - main.py
- controller
 - agent.py
- state
 - state.py
- core

openhands/agenthub/browsing_agent/browsing_agent.py

```
@@ -198,7 +198,7 @@ def step(self, state: State) -> Action:
198 198         except Exception as e:
199 199             logger.error(
200 200                 'Error when trying to process the accessibility tree: %s', e
201 -             )
201 +             , exc_info=True)
202 202         return MessageAction('Error encountered when browsing.')
203 203
204 204     goal, _ = state.get_current_user_intent()
```

openhands/agenthub/browsing_agent/response_parser.py

```
@@ -110,7 +110,7 @@ def parse(self, action_str: str) -> Action:
110 110         args = tree.body[0].value.args # type: ignore
111 111         msg_content = args[0].value
112 112         except SyntaxError:
113 -             logger.error(f'Error parsing action: {sub_action}')
113 +             logger.error(f'Error parsing action: {sub_action}', exc_info=True)
114 114         # the syntax was not correct, but we can still try to get the message
115 115         # e.g. send_msg_to_user("Hello, world!") or send_msg_to_user('Hello, wor
116 116         match = re.search(r'send_msg_to_user\(((["\'])(.*?)\1)\)', sub_action)
```

Dev-o-calypse

Dev-o-calypse

aka

Innovator's Paradise

Abstraction Leap

Abstraction Leap

Prompts-as-code

Natural language becomes the human-facing artifact

Structured and tested... somehow

Objection: Predictability

Abstraction Leap

Prompts-as-code

```
parse(program): Read a Scheme expression from a string.
```

```
'(1 + (2 * 3))' -> [1, '+', [2, '*', 3]]
```

```
tokenize(s): Convert a string into a list of tokens, including parens.
```

```
"1 + 2" -> ['1', '+', '2']
```

```
"1 + (2 * 3)" -> ['1', '+', '(', '2', '*', '3', ')']
```

61 lines of prompts and examples

-> 220 line Lisp interpreter in Python

Parsel 2022 paper, github.com/ezelikman/parasel

Abstraction Leap

Domain-Specific Languages (DSLs)

Abstraction Leap

Domain-Specific Languages (DSLs)

Invest in a specialized programming environment

Language workbenches + LLMs improve cost tradeoff

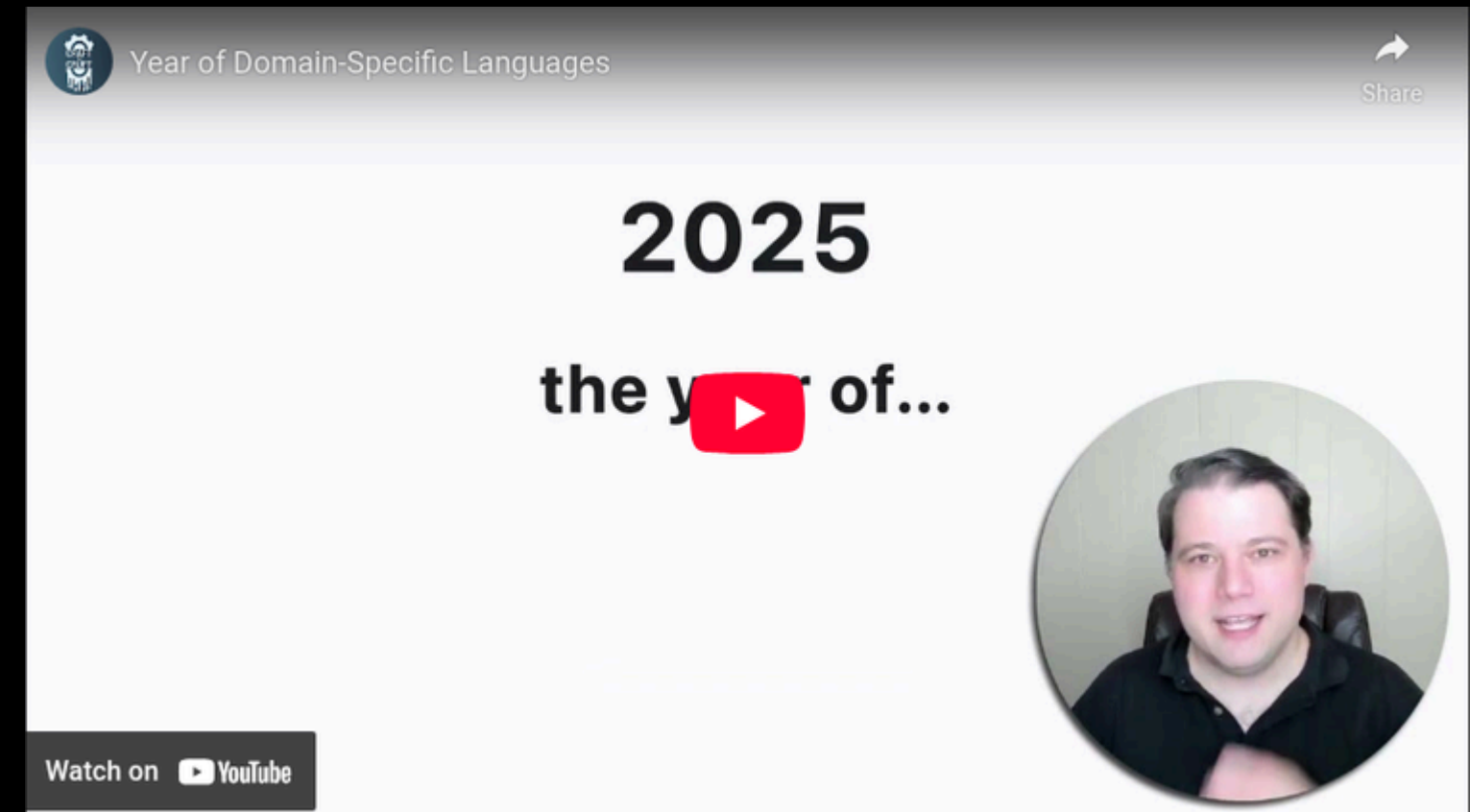
See: JetBrains MPS, Racket, LionWeb, Xtext

Abstraction Leap

Domain-Specific Languages (DSLs)



Empowerment of Subject-Matter Experts
Markus Voelter



Year of DSLs
Craft vs Cruft (me)

Uncharted Waters



Uncharted Waters

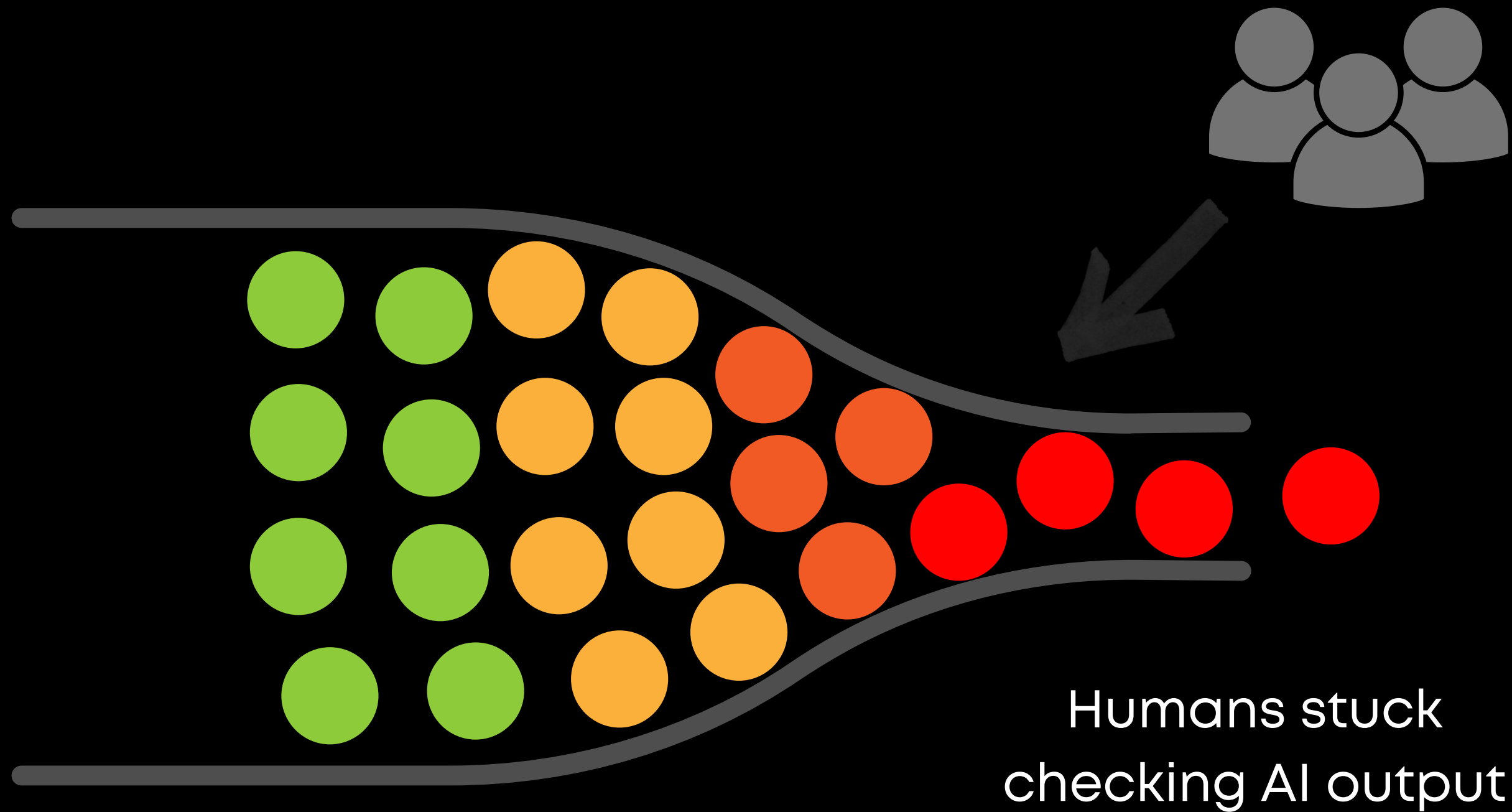
Future of computing isn't code-like

Direct model inference?

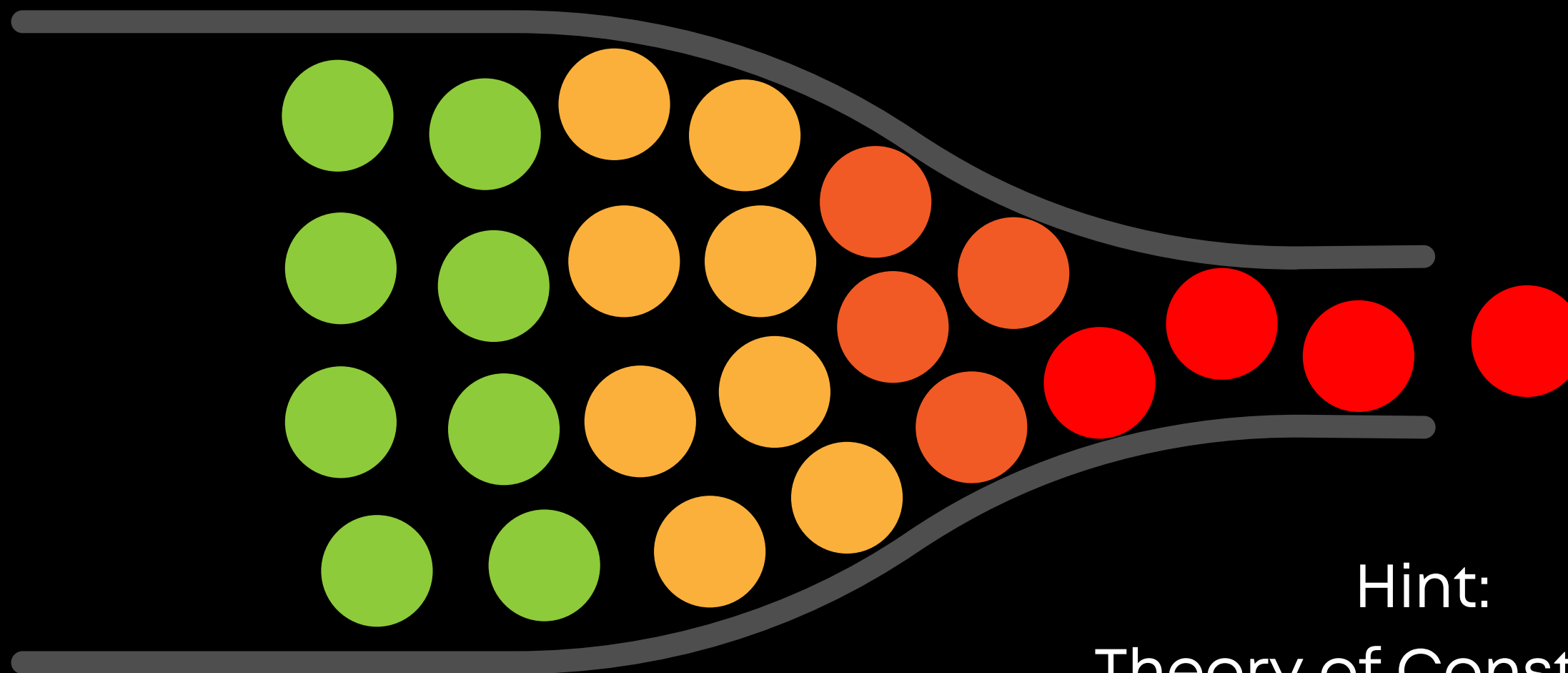
AI invents new paradigms?

Review Economy

Review Economy



Review Economy



Hint:
Theory of Constraints

Infinite Pile of Garbage



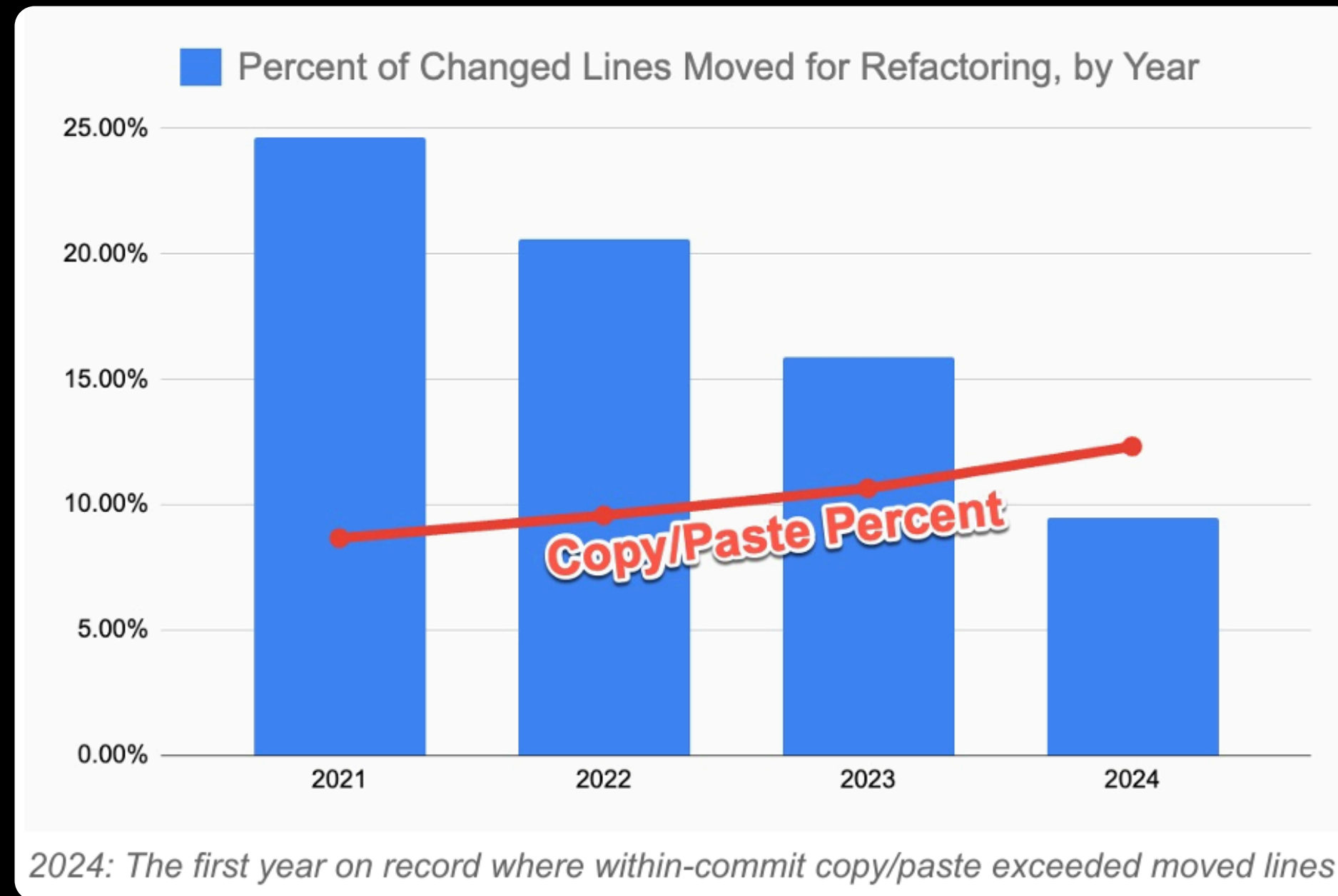
Infinite Pile of Garbage

“Developers with Copilot access saw a **significantly higher bug rate [41%]** while their issue throughput remained consistent”

Source: Uplevel Whitepaper

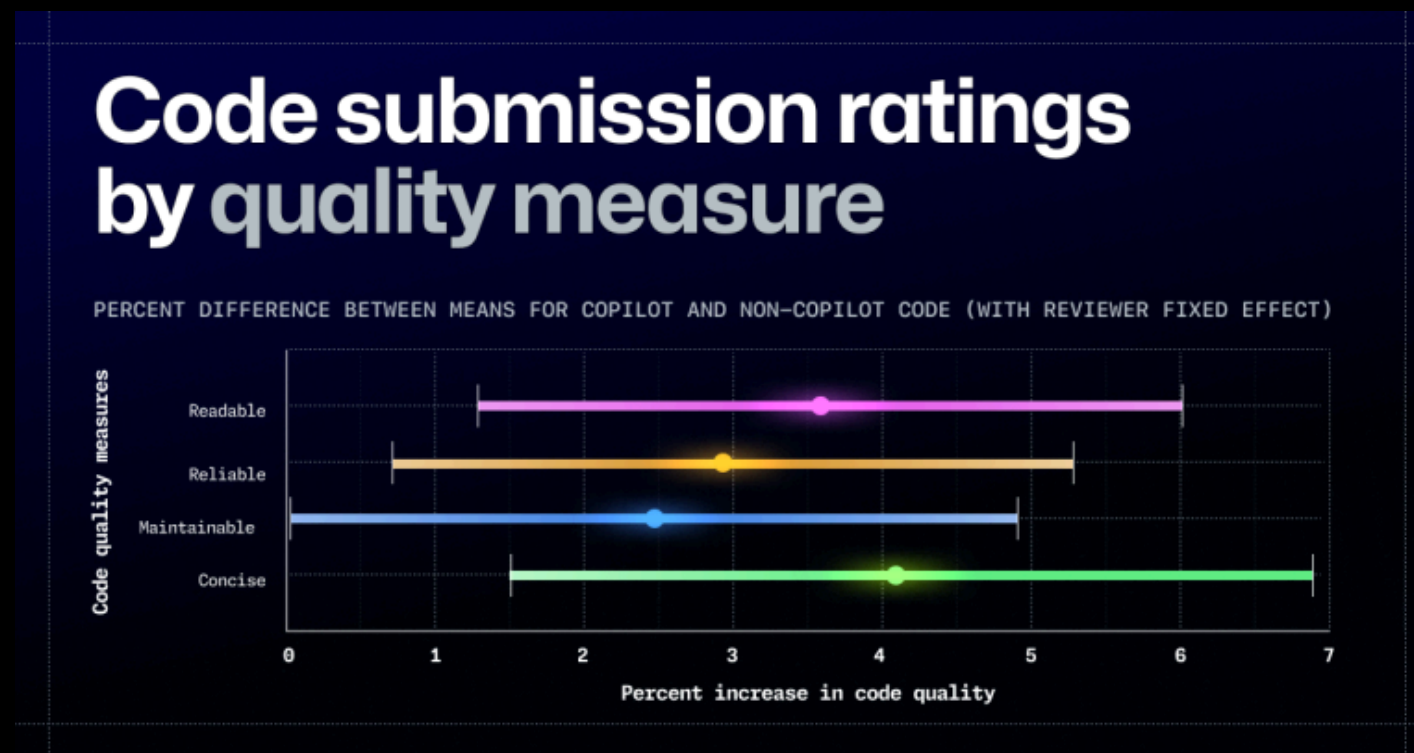
“Can Generative AI Improve Developer Productivity?”

Infinite Pile of Garbage



Source: GitClear whitepaper
“AI Copilot Code Quality”

Infinite Pile of Garbage



Rebuttal: GitHub found Copilot increased quality

Source: GitHub Blog

“Does GitHub Copilot improve code quality?
Here’s what the data says”

Infinite Pile of Garbage

It will get better...
if we make it better

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2. ~~Dev-o-calypse~~

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Multiple endgames

We get a say

What is the goal?

Thank you